

Assignment 4 Report

Optimizing Transformer Translation using Ray Tune & Optuna

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Part 1 — Baseline Metrics

The baseline Transformer model was trained using the provided notebook without any modifications.

Metric	Value
Epochs	100
Training Time	128 minutes
Final Training Loss	0.0972
BLEU Score	52.47

Table 1: Baseline Performance

Part 2 — Hyperparameter Tuning

The training loop was refactored into a `train_tune(config)` function compatible with Ray Tune.

2.1 Hyperparameters Tuned

Hyperparameter	Range	Method
Learning Rate	10^{-5} to 10^{-3}	loguniform
Batch Size	32, 64, 128	choice
Attention Heads	4, 8	choice
FeedForward Dim	1024, 2048, 4096	choice
Dropout Rate	0.05 to 0.35	uniform
Number of Layers	3, 4, 6	choice

Table 2: Hyperparameter Search Space

2.2 Search Configuration

- Search Algorithm: Optuna TPE

- Number of Trials: 12
- Scheduler: ASHA (early stopping)
- Max Epochs per Trial: 30

Part 3 — Best Configuration Found

Hyperparameter	Best Value
Learning Rate	0.000371
Batch Size	64
Attention Heads	4
FeedForward Dimension	1024
Dropout Rate	0.065
Number of Layers	3

Table 3: Best Configuration

Observation: A smaller model (3 layers) performed better on the dataset. Reduced depth helped faster convergence and improved generalization.

Part 4 — Final Results

Metric	Baseline	Tuned Model	Improvement
Epochs	100	30	Reduced
BLEU Score	52.47	71.47	+19.00
Final Loss	0.0972	1.30	Increased
Training Time	128 min	14.7 min	$\sim 9\times$ faster

Table 4: Final Comparison

Note: The higher loss value in the tuned model is expected due to the use of label smoothing, which prevents overconfident predictions.